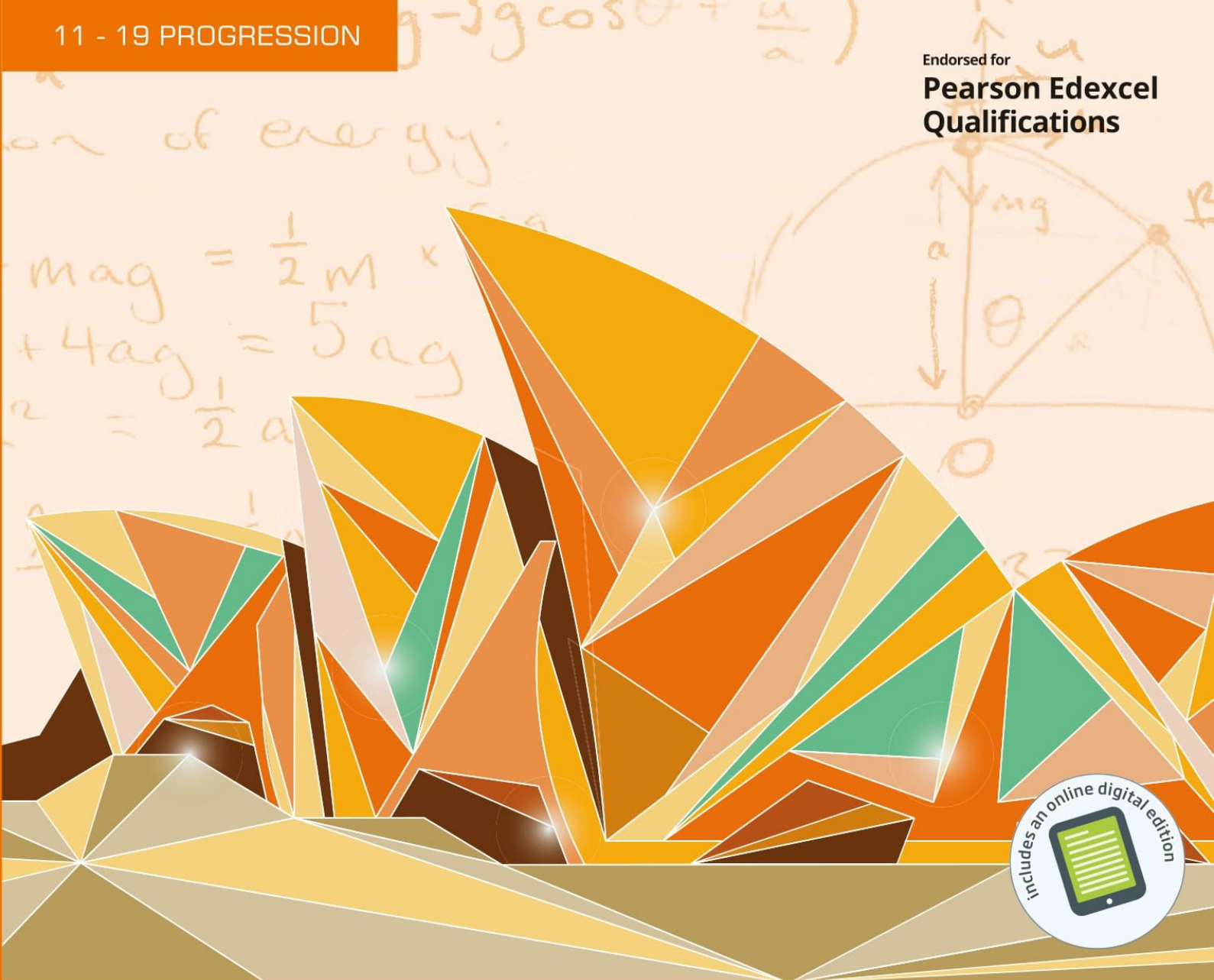




Pearson Edexcel AS and A level Further Mathematics

Further Mechanics 1

FM1

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● = A level only

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Overarching themes

The following three overarching themes have been fully integrated throughout the Pearson Edexcel AS and A level Mathematics series, so they can be applied alongside your learning and practice.

1. Mathematical argument, language and proof

- Rigorous and consistent approach throughout
- Notation boxes explain key mathematical language and symbols
- Dedicated sections on mathematical proof explain key principles and strategies
- Opportunities to critique arguments and justify methods

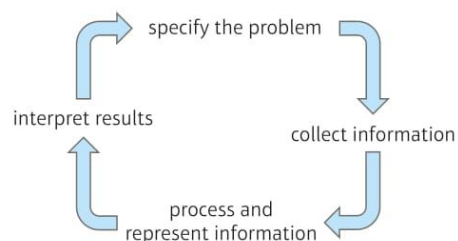
2. Mathematical problem solving

- Hundreds of problem-solving questions, fully integrated into the main exercises
- Problem-solving boxes provide tips and strategies
- Structured and unstructured questions to build confidence
- Challenge boxes provide extra stretch

3. Mathematical modelling

- Dedicated modelling sections in relevant topics provide plenty of practice where you need it
- Examples and exercises include qualitative questions that allow you to interpret answers in the context of the model
- Dedicated chapter in Statistics & Mechanics Year 1/AS explains the principles of modelling in mechanics

The Mathematical Problem-solving cycle



Finding your way around the book

Access an online digital edition using the code at the front of the book.



Each chapter starts with a list of objectives

The real world applications of the maths you are about to learn are highlighted at the start of the chapter with links to relevant questions in the chapter



The *Prior knowledge check* helps make sure you are ready to start the chapter

A level content is clearly flagged

Exercises are packed with exam-style questions to ensure you are ready for the exams

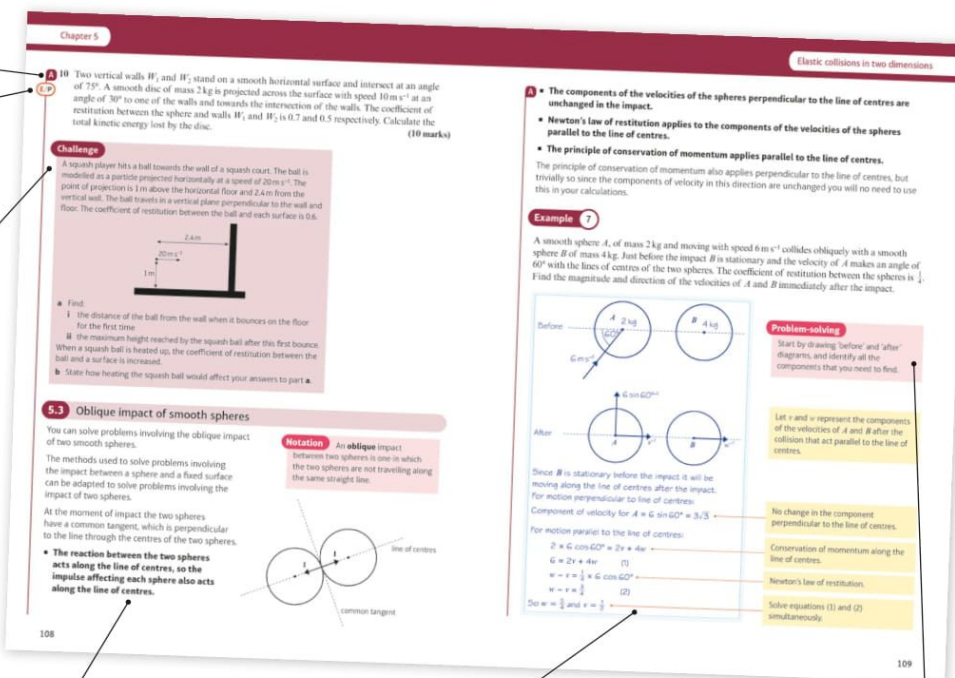
Challenge boxes give you a chance to tackle some more difficult questions

Exercise questions are carefully graded so they increase in difficulty and gradually bring you up to exam standard

Exam-style questions are flagged with **E**

Problem-solving questions are flagged with **P**

Every few chapters a *Review exercise* helps you consolidate your learning with lots of exam-style questions

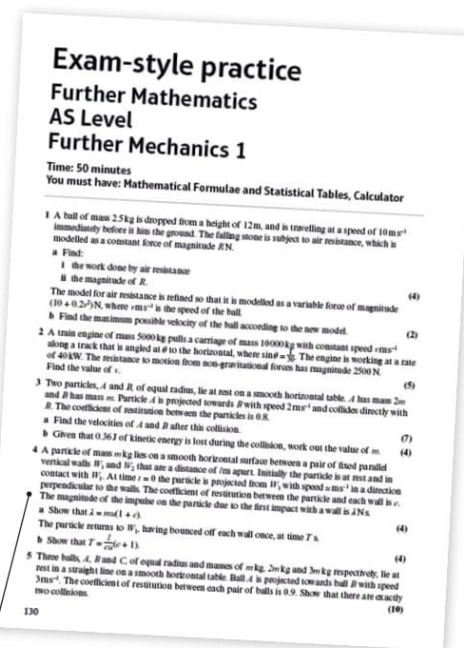
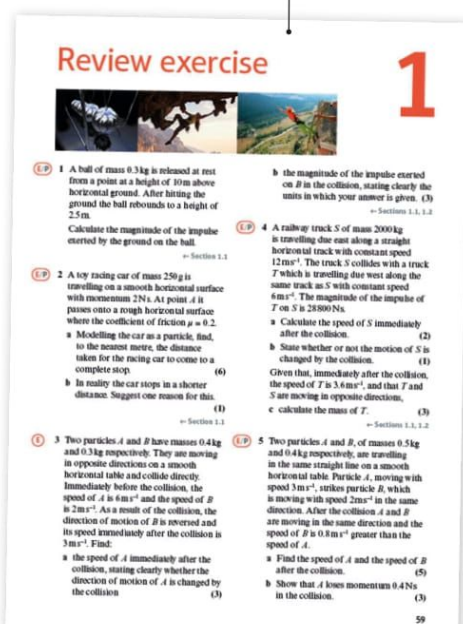


Each section begins with explanation and key learning points

Each chapter ends with a *Mixed exercise* and a *Summary of key points*

Step-by-step worked examples focus on the key types of questions you'll need to tackle

Problem-solving boxes provide hints, tips and strategies, and *Watch out* boxes highlight areas where students often lose marks in their exams



AS and A level practice papers at the back of the book help you prepare for the real thing.

Momentum and impulse

1

Objectives

After completing this chapter you should be able to:

- Calculate the momentum of a particle and the impulse of a force → pages 2–4
- Solve problems involving collisions using the principle of conservation of momentum → pages 4–9
- Use the impulse–momentum principle and the principle of conservation of momentum in vector form → pages 9–12

Prior knowledge check

- 1 The forces $\mathbf{F}_1 = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{F}_2 = 5\mathbf{i} + 4\mathbf{j}$ act on a particle. Find the magnitude and direction of the resultant force.
← Statistics and Mechanics 1, Chapter 10
- 2 A particle moves in a straight line with constant acceleration.
Given s = displacement in m, u = initial velocity in m s^{-1} , v = final velocity in m s^{-1} , a = acceleration in m s^{-2} and t = time in seconds, find:
a v when $u = 3$, $a = 0.5$, $t = 5$
b s when $u = 4.5$, $a = -1.5$, $t = 2$
← Statistics and Mechanics 1, Chapter 9
- 3 A body of mass 2 kg is acted on by a force $\mathbf{F}$ N. The body starts from rest and moves in a straight line. After 5 seconds, the displacement of the body is 20 m. Find the magnitude of $\mathbf{F}$.
← Statistics and Mechanics 1, Chapters 9, 10

Newton's cradle demonstrates the **principle of conservation of momentum**. When the first ball collides with the second, the first ball stops, but its momentum is transferred to the second ball, then the third, then the fourth, until it reaches the very last ball.

1.1 Momentum in one dimension

You can calculate the momentum of a particle and the impulse of a force.

- **The momentum of a body of mass m which is moving with velocity v is mv .**

If m is in kg and v is in m s^{-1} then the units of momentum will be kg m s^{-1} .

However, since $\text{kg m s}^{-1} = (\text{kg m s}^{-2}) \text{ s}$ and kg m s^{-2} are the units for force ($F = ma$) you can also measure momentum in newton seconds (N s).

Note Velocity is a vector quantity and mass is a scalar, so momentum is a vector quantity.

Example 1

Find the magnitude of the momentum of:

- a cricket ball of mass 400 g moving at 18 m s^{-1}
- a lorry of mass 5 tonnes moving at 0.3 m s^{-1} .

a Momentum = mass $\times$ velocity

$$\begin{aligned}\text{Magnitude of momentum} &= \frac{400}{1000} \times 18 \\ &= 7.2 \text{ N s}\end{aligned}$$

The mass must be in kg.

The units can be N s or kg m s^{-1} .

b Momentum = mass $\times$ velocity

$$\begin{aligned}\text{Magnitude of momentum} &= (5 \times 1000) \times 0.3 \\ &= 1500 \text{ kg m s}^{-1}\end{aligned}$$

- **If a constant force F acts for time t then we define the impulse of the force to be Ft .**

If F is in N and t is in s then the units of impulse will be N s.

Note Force is a vector quantity and time is a scalar, so impulse is a vector quantity.

Examples of an impulse include a bat hitting a ball, a snooker ball hitting another ball or a jerk in a string when it suddenly goes tight. In all these cases the time for which the force acts is very small but the force is quite large and so the product of the two, which gives the impulse, is of reasonable size. However, there is no theoretical limit on the size of t .

Suppose a body of mass m is moving with an initial velocity u and is then acted upon by a force F for time t . This results in its final velocity being v .

Its acceleration is given by $a = \frac{v - u}{t}$

Substituting into $F = ma$: $F = m\left(\frac{v - u}{t}\right)$

$$\begin{aligned}Ft &= m(v - u) \\ &= mv - mu\end{aligned}$$

The impulse of the force I is given by $I = Ft$.

- **$I = mv - mu$**

Impulse = final momentum – initial momentum

Impulse = change in momentum

This is called the **impulse–momentum principle**.

Watch out This is a vector equation, so for motion in a straight line a positive direction must be chosen and each value must be given the correct sign.

Example 2

A body of mass 2 kg is initially at rest on a smooth horizontal plane. A horizontal force of magnitude 4.5 N acts on the body for 6 s. Find:

- the magnitude of the impulse given to the body by the force
- the final speed of the body.

$$\begin{aligned} \text{a Magnitude of the impulse} &= \text{force} \times \text{time} \\ &= 4.5 \times 6 \\ &= 27 \text{ N s} \end{aligned}$$

The units can be N s or kg m s^{-1} .

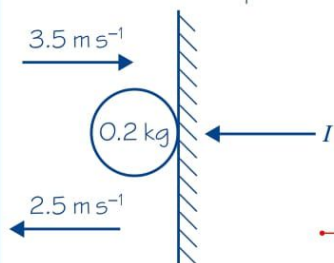
$$\begin{aligned} \text{b Impulse} &= \text{Final momentum} - \text{Initial momentum} \\ 27 &= 2v - 0 \\ v &= 13.5 \text{ m s}^{-1} \end{aligned}$$

The body is at rest initially.

Example 3

A ball of mass 0.2 kg hits a fixed vertical wall at right angles with speed 3.5 m s^{-1} . The ball rebounds with speed 2.5 m s^{-1} . Find the magnitude of the impulse exerted on the wall by the ball.

This diagram shows the initial and final velocities of the ball and the impulse acting on it.



$$\begin{aligned} (\leftarrow): I &= (0.2 \times 2.5) - (0.2 \times (-3.5)) \\ &= 0.5 + 0.7 \\ &= 1.2 \text{ N s} \end{aligned}$$

Therefore, by Newton's 3rd law, the magnitude of the impulse exerted on the wall by the ball is 1.2 N s.

Problem-solving

Because the wall is fixed you cannot apply the impulse-momentum principle to it. Find the magnitude of the impulse exerted on the ball by the wall and then use Newton's 3rd law to deduce that the magnitude of the impulse exerted on the wall by the ball will be the same.

Note that this is a plan view of the situation.

Choose a positive direction ($\leftarrow$) and apply the impulse-momentum principle to the ball.

The initial velocity is in the negative direction.

Exercise 1A

- A ball of mass 0.5 kg is at rest when it is struck by a bat and receives an impulse of 15 N s. Find its speed immediately after it is struck.
- A ball of mass 0.3 kg moving along a horizontal surface hits a fixed vertical wall at right angles with speed 3.5 m s^{-1} . The ball rebounds at right angles to the wall. Given that the magnitude of the impulse exerted on the ball by the wall is 1.8 N s, find the speed of the ball just after it rebounds.

3 A toy car of mass 0.2 kg is pushed from rest along a smooth horizontal floor by a horizontal force of magnitude 0.4 N for 1.5 s . Find its speed at the end of the 1.5 s .

E 4 A ball of mass 0.2 kg , moving along a horizontal surface, hits a fixed vertical wall at right angles. The ball rebounds at right angles to the wall with speed 3.5 m s^{-1} . Given that the magnitude of the impulse exerted on the ball by the wall is 2 N s , find the speed of the ball just before it hits the wall. **(3 marks)**

E/P 5 A ball of mass 0.2 kg is dropped from a height of 2.5 m above horizontal ground. After hitting the ground it bounces to a height of 1.8 m above the ground. Find the magnitude of the impulse received by the ball from the ground. **(4 marks)**

1.2 Conservation of momentum

You can solve problems involving collisions using the principle of conservation of momentum.

By Newton's 3rd law, when two bodies collide, each one exerts an equal and opposite force on the other. They are in contact for the same time, so they each exert an impulse on the other of equal magnitude but in opposite directions.

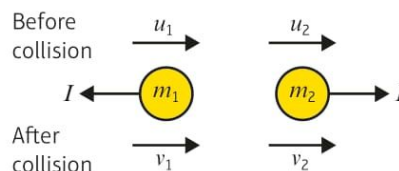


By the impulse–momentum principle the changes in momentum of each body are equal but opposite in direction. Thus, these changes in momentum cancel each other out, and the momentum of the whole system is unchanged. This is called the principle of **conservation of momentum**.

■ Total momentum before impact = total momentum after impact

You can write this in symbols for two masses m_1 and m_2 with velocities u_1 and u_2 respectively before the collision, and velocities v_1 and v_2 respectively after the collision:

$$\mathbf{m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2}$$



When solving problems involving collisions, always:

- draw a diagram showing the velocities before and after the collision with arrows
- if appropriate, include the impulses on your diagram with arrows
- choose a positive direction and apply the impulse–momentum principle and/or the principle of conservation of momentum.

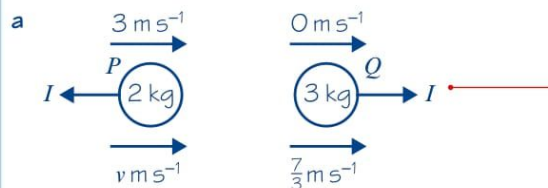
Example 4

A particle P of mass 2 kg is moving with speed 3 m s^{-1} on a smooth horizontal plane. Particle Q of mass 3 kg is at rest on the plane. Particle P collides with particle Q and after the collision Q moves off with speed $\frac{7}{3}\text{ m s}^{-1}$. Find:

- the speed and direction of motion of P after the collision
- the magnitude of the impulse received by P in the collision.

Online Explore particle collisions using GeoGebra.





Conservation of momentum: ($\rightarrow$)

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$(2 \times 3) + (3 \times 0) = 2v + \left(3 \times \frac{7}{3}\right)$$

$$6 = 2v + 7$$

$$-\frac{1}{2} = v$$

The direction of motion of P is reversed by the collision and its speed is $\frac{1}{2} \text{ m s}^{-1}$.

b For Q : ($\rightarrow$) $I = 3\left(\frac{7}{3} - 0\right)$
 $= 7 \text{ N s}$

Alternatively, for P : ($\leftarrow$)

$$I = 2((-v) - (-3))$$

$$= 2\left(\frac{1}{2} + 3\right)$$

$$= 7 \text{ N s}$$

So the impulse received by P has magnitude 7 N s .

Draw a diagram showing the velocities before and after the collision (with arrows) and the impulses (with arrows).

Choose a positive direction and apply the principle of conservation of momentum.

Since v is negative, P moves in the opposite direction after the collision.

Watch out The direction of motion of P in your answer must be with reference to the original direction of motion of P . Do not use the words left or right.

To find the impulse consider one particle and apply the impulse-momentum principle. Here it is easier to consider Q .

Since each particle receives an impulse of equal magnitude, the magnitude of the impulse received by P is also 7 N s .

Example 5

Two particles A and B of masses 2 kg and 4 kg respectively are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide. Before the collision the speeds of A and B are 3 m s^{-1} and 2 m s^{-1} respectively. After the collision the direction of motion of A is reversed and its speed is 2 m s^{-1} . Find:

- the speed and direction of B after the collision
- the magnitude of the impulse given by A to B in the collision.



Conservation of momentum: ($\rightarrow$)

$$(2 \times 3) + (4 \times (-2)) = (2 \times (-2)) + 4v$$

$$6 - 8 = -4 + 4v$$

$$2 = 4v$$

$$0.5 = v$$

B has speed 0.5 m s^{-1} and its direction of motion is reversed by the collision.

Online Explore collisions with two moving particles using GeoGebra.

You need to 'guess' the direction of B after the collision. If it is moving in the other direction the answer will be negative.

This defines the positive direction.

Each velocity must be given the correct sign.

As the value of v is positive the 'guess' for the direction of B after the collision was correct.

- b For A : ($\leftarrow$)
impulse–momentum principle

$$I = 2(2 - (-3))$$

$$= 10 \text{ N s}$$

The magnitude of the impulse given by A to B is 10 N s .

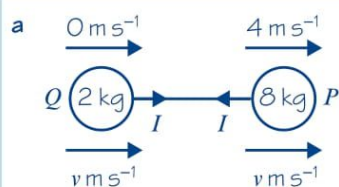
Although we could consider either particle, it is safer to consider A since its initial and final speed were given in the question.

The magnitude of the impulse given by A to B is the same as the magnitude of the impulse given by B to A .

Example 6

Two particles P and Q , of masses 8 kg and 2 kg respectively, are connected by a light inextensible string. The particles are at rest on a smooth horizontal plane with the string slack. Particle P is projected directly away from Q with speed 4 m s^{-1} .

- Find the common speed of the particles after the string goes taut.
- Find the magnitude of the impulse transmitted through the string when it goes taut.



Using conservation of momentum ($\rightarrow$):

$$(2 \times 0) + (8 \times 4) = 2v + 8v$$

$$32 = 10v$$

$$3.2 = v$$

The common speed of the particles is 3.2 m s^{-1} .

- b For Q ($\rightarrow$):

$$I = 2(v - 0)$$

$$= 2 \times 3.2$$

$$= 6.4 \text{ N s}$$

The magnitude of the impulse transmitted through the string (the 'jerk') is 6.4 N s .

The string is inextensible so these are the same.

This must be applied to the whole system.

To find the impulse we must consider one of the particles and apply the impulse–momentum principle. It is easier to consider Q .

Example 7

Two particles A and B of masses 2 kg and 4 kg respectively are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide. Before the collision the speeds of A and B are 3 m s^{-1} and 2 m s^{-1} respectively. Given that the magnitude of the impulse due to the collision is 7 N s , find:

- the velocity of A after the collision
- the velocity of B after the collision.

For A : ($\leftarrow$)

$$7 = 2(v_1 - (-3))$$

$$7 = 2(v_1 + 3)$$

$$3.5 = v_1 + 3$$

$$0.5 = v_1$$

For B : ($\rightarrow$)

$$7 = 4(v_2 - (-2))$$

$$1.75 = v_2 + 2$$

$$-0.25 = v_2$$

a The direction of motion of A is reversed and its speed is 0.5 m s^{-1} .

b The direction of motion of B is unchanged and its speed is 0.25 m s^{-1} .

Online Explore particle collisions with known impulse using GeoGebra.



The diagram should show the velocities and impulses with arrows.

Again you need to guess which way the particles go after the collision. There are three sensible possibilities: the one shown, $\overleftarrow{v_1} \overrightarrow{v_2}$ or $\overleftarrow{v_1} \overleftarrow{v_2}$.

To find either v_1 or v_2 consider one particle only, choose a positive direction and apply the impulse-momentum principle.

As v_1 is positive, the guess for the direction of A was correct.

As v_2 is negative, the guess for the direction of B was incorrect and it travels in the opposite direction to that shown on the diagram.

Watch out The question asks for the **velocities** so you need to state the speed and the direction of motion.

Exercise 1B

- A particle P of mass 2 kg is moving on a smooth horizontal plane with speed 4 m s^{-1} . It collides with a second particle Q of mass 1 kg which is at rest. After the collision P has speed 2 m s^{-1} and it continues to move in the same direction. Find the speed of Q after the collision.
- A railway truck of mass 25 tonnes moving at 4 m s^{-1} collides with a stationary truck of mass 20 tonnes . As a result of the collision the trucks couple together. Find the common speed of the trucks after the collision.
- Particles A and B have masses 0.5 kg and 0.2 kg respectively. They are moving with speeds 5 m s^{-1} and 2 m s^{-1} respectively in the same direction along the same straight line on a smooth horizontal surface when they collide. After the collision A continues to move in the same direction with speed 4 m s^{-1} . Find the speed of B after the collision.
- A particle of mass 2 kg is moving on a smooth horizontal plane with speed 4 m s^{-1} . It collides with a second particle of mass 1 kg which is at rest. After the collision the particles join together.
 - Find the common speed of the particles after the collision.
 - Find the magnitude of the impulse in the collision.
- Two particles A and B of masses 2 kg and 5 kg respectively are moving towards each other along the same straight line on a smooth horizontal surface. The particles collide. Before the collision the speeds of A and B are 6 m s^{-1} and 4 m s^{-1} respectively. After the collision the direction of motion of A is reversed and its speed is 1.5 m s^{-1} . Find:
 - the speed and direction of B after the collision (3 marks)
 - the magnitude of the impulse given by A to B in the collision. (3 marks)

- a the speed of P immediately after the collision (3 marks)
 b the magnitude of the impulse exerted by P on Q during the collision. (3 marks)

E 6 A particle P of mass 1.5 kg is moving along a straight horizontal line with speed 3 m s^{-1} . Another particle Q of mass 2.5 kg is moving, in the opposite direction, along the same straight line with speed 4 m s^{-1} . The particles collide. Immediately after the collision the direction of motion of P is reversed and its speed is 2.5 m s^{-1} .

- a Calculate the speed of Q immediately after the impact. (3 marks)
 b State whether or not the direction of motion of Q is changed by the collision. (1 mark)
 c Calculate the magnitude of the impulse exerted by Q on P , giving the units of your answer. (3 marks)

E/P 7 A particle A of mass m is moving with speed $2u$ in a straight line on a smooth horizontal table. It collides with another particle B of mass km which is moving in the same straight line on the table with speed u in the opposite direction to A . In the collision, the particles form a single particle which moves with speed $\frac{2}{3}u$ in the original direction of A 's motion. Find the value of k . (3 marks)

E/P 8 A metal pin of mass 2 kg is driven vertically into the ground by a blow from a sledgehammer of mass 10 kg . The hammer falls vertically on to the pin, its speed just before impact being 9 m s^{-1} . In a model of the situation it is assumed that, after impact, the pin and the hammer stay in contact and move together before coming to rest.

- a Find the speed of the pin immediately after impact. (3 marks)
 The pin moves 3 cm into the ground before coming to rest. Assuming in this model that the ground exerts a constant resistive force of magnitude R newtons as the pin is driven down,
 b find the value of R . (5 marks)
 c State one way in which this model might be refined to be more realistic. (1 mark)

A
E 9 A cricket ball of mass 0.5 kg is struck by a bat. Immediately before being struck the velocity of the ball is $-25\mathbf{i}\text{ m s}^{-1}$. Immediately after being struck the velocity of the ball is $(23\mathbf{i} + 20\mathbf{j})\text{ m s}^{-1}$. Find the magnitude of the impulse exerted on the ball by the bat and the angle between the impulse and the direction of $\mathbf{i}$. (5 marks)

E 10 A ball of mass 0.2 kg is hit by a bat which gives it an impulse of $(2.4\mathbf{i} + 3.6\mathbf{j})\text{ N s}$. The velocity of the ball immediately after being hit is $(12\mathbf{i} + 5\mathbf{j})\text{ m s}^{-1}$. Find the velocity of the ball immediately before it is hit. (3 marks)

E 11 A body P of mass 4 kg is moving with velocity $(2\mathbf{i} + 16\mathbf{j})\text{ m s}^{-1}$ when it collides with a body Q of mass 3 kg moving with velocity $(-\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$. Immediately after the collision the velocity of P is $(-4\mathbf{i} - 32\mathbf{j})\text{ m s}^{-1}$. Find the velocity of Q immediately after the collision. (3 marks)

E/P 12 A particle P of mass 0.3 kg is moving so that its position vector $\mathbf{r}$ metres at time t seconds is given by

$$\mathbf{r} = (t^3 + t^2 + 4t)\mathbf{i} + (11t)\mathbf{j}, \quad t \leq 4.$$

- a Calculate the speed of P when $t = 4$. (4 marks)
 When $t = 4$, the particle is given an impulse $(2.4\mathbf{i} + 3.6\mathbf{j})\text{ N s}$.
 b Find the velocity of P immediately after the impulse. (3 marks)

Pearson Edexcel AS and A level Further Mathematics

Further Mechanics 1

Series Editor: Harry Smith

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Year 1/AS

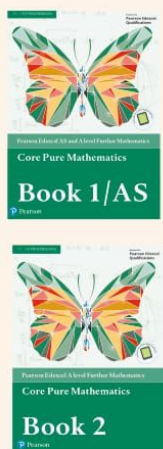


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